Trustworthy Traffic Signal Control with Deep Reinforcement Learning

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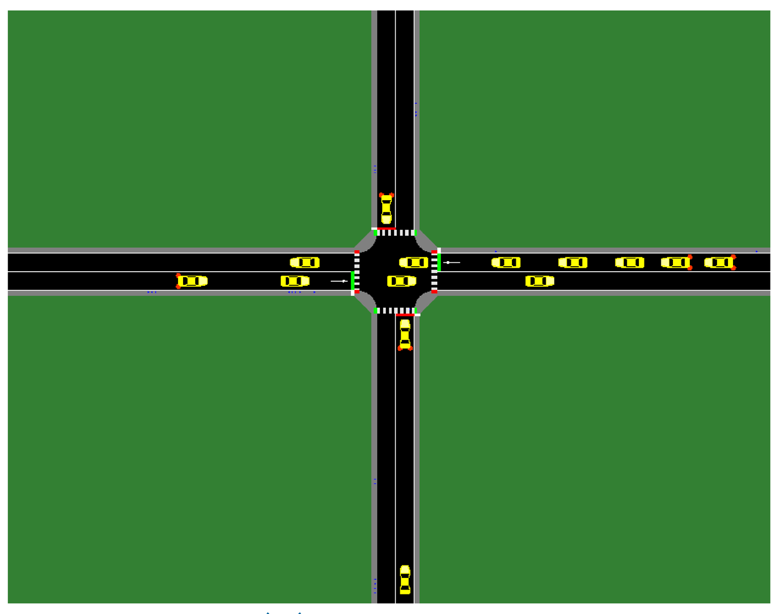
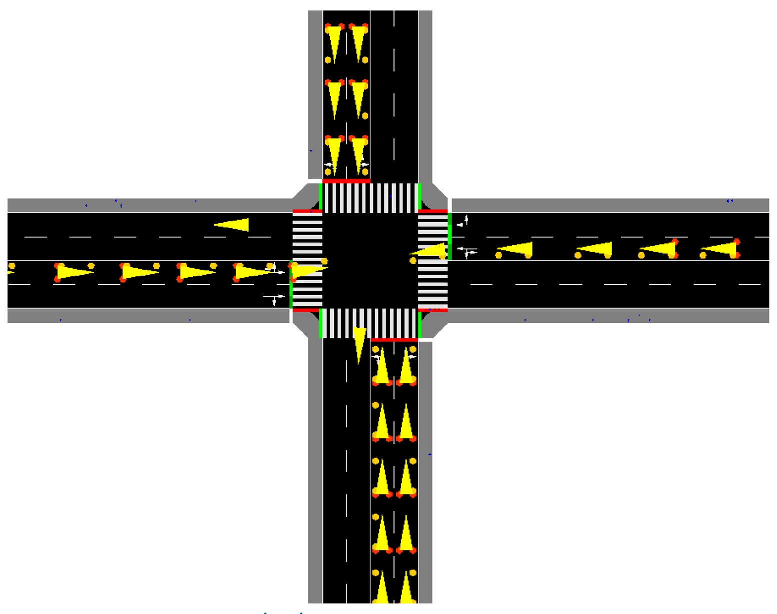
Abstract

Traffic signal control in urban environments requires balancing vehicle throughput and pedestrian safety. To address this challenge, we propose an **Action-Constrained Proximal Policy Optimisation (PPO)** framework that integrates pedestrian safety requirements through feasible-action masking during policy execution. The proposed method is compared with five widely used traffic-control baselines — Fixed-Time, Max Pressure, SOTL, DQN-AR, and SPRe+ — in both a single-intersection task and a 3×3 grid network simulated in SUMO. Through these experiments, we examine the potential of safe and explainable deep reinforcement learning for trustworthy traffic signal control in smart-city environments.



(a) 1x1 env



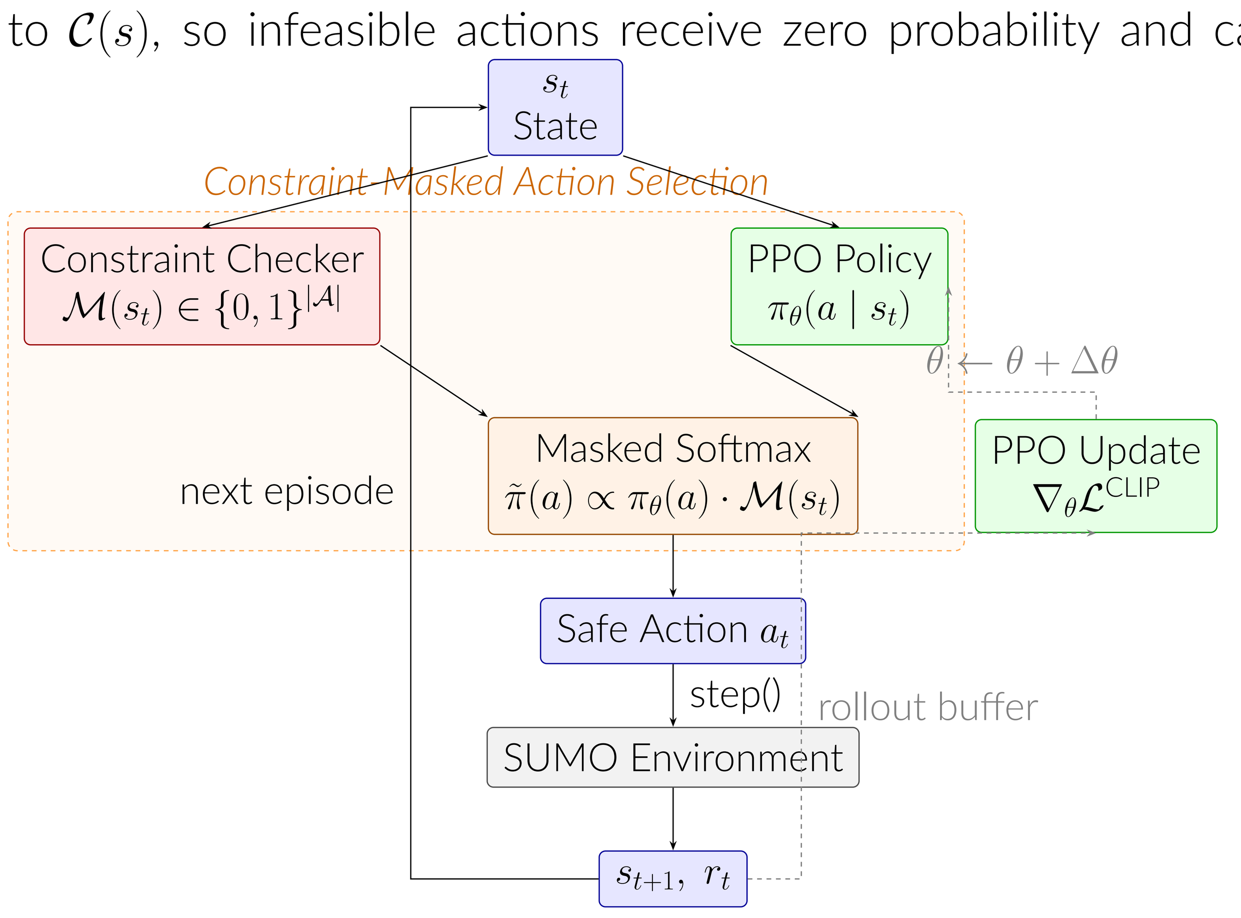
(b) 3x3 env

Research Objectives

- **Objective 1:** Design an action-constrained RL agent that enforces pedestrian safety constraints *by construction*, guaranteeing zero violations without relying on reward shaping alone.
- **Objective 2:** Benchmark the proposed agent against rule-based baselines (Fixed-Time, Max Pressure, SOTL) and action-constrained learned agents (DQN-AR, SPRe+) on efficiency and safety metrics across two network scales.
- **Objective 3:** Validate the computational advantage of acceptance-rejection masking over QP-projection (SPRe+) in growing action spaces.

Methodology

The agent operates within an **Action-Constrained MDP**. At every step, the constraint checker computes the feasible action set $\mathcal{C}(s)$ by removing phases that violate any hard constraint. PPO selects actions via a **masked softmax** restricted to $\mathcal{C}(s)$, so infeasible actions receive zero probability and can never be executed.



Experimental Setup

Simulation Scenarios

Parameter	1×1 Scenario	3×3 Scenario
Network topology	Single 4-way intersection	3×3 grid
Controlled TLS	1	B1 (center node)
Simulation length	3 600 s	3 600 s
Training episodes	100	100
Evaluation episodes	5	5
State dimension	21	31
Action space	2 phases	4 phases

Agents & Baselines

- **Fixed-Time:** Fixed 30 s NS/EW cycle. No traffic awareness.
- **Max Pressure:** Selects phase maximising queue pressure. Vehicle-centric; no pedestrian safety.
- **SOTL:** Switches when red-side queue $\Phi_{\text{red}}(t) \geq \kappa = 5$ and $\tau \geq \tau_{\text{min}}$.
- **DQN-AR:** DQN with acceptance-rejection safety filter. Zero violations; $\mathcal{O}(1)$ per-step cost.
- **SPRe+ :** Projects policy onto feasible simplex via QP. $\mathcal{O}(|\mathcal{A}|^2)$; equivalent to DQN-AR here.
- **Action-Constrained PPO (proposed):** Masked softmax over $\mathcal{C}(s)$. Learns safe behaviour during training.

Results & Discussion

Table 1 — 1×1 Single Intersection (5 episodes × 3 600 s)

Agent	Avg Wait (s)	Avg Queue	Viol. Rate	Reward
Action-Constr. PPO	**60.28**	**4.27**	**0.0%**	**−217,026**
DQN-AR	65.85	4.47	0.0%	−237,073
SPRe+ (DQN policy)	65.85	4.47	0.0%	−237,073
SOTL ($\kappa = 5$)	71.64	4.78	18.6%	−291,292
Max Pressure†	55.63	4.04	53.5%	−297,984
Fixed-Time (30 s)	107.58	5.61	57.2%	−490,231

†Lowest raw wait but 53.5% violation rate — unsafe for real deployment.

Table 2 — 3×3 Grid, Intersection B1 (5 episodes × 3 600 s)

Agent	Avg Wait (s)	Avg Queue	Viol. Rate	Reward
Action-Constr. PPO	**1,291.4**	**78.67**	**0.0%**	**−4,649,048**
DQN-AR	1,350.6	79.61	0.0%	−4,862,170
SPRe+ (DQN policy)	1,350.6	79.61	0.0%	−4,862,170
Max Pressure	1,380.1	79.58	55.8%	−5,068,558
SOTL ($\kappa = 5$)	1,406.0	80.65	71.0%	−5,189,343
Fixed-Time (30 s)	1,604.2	80.27	69.1%	−5,899,362

Key Findings

- **High Efficiency**: PPO reduces vehicle wait times by **8.46%** (1×1) and **4.38%** (3×3) compared to DQN-AR.
- **Guaranteed Safety**: Learned agents achieve **0% violations**. In contrast, rule-based heuristics violate safety rules in **19–71%** of steps (e.g., Max Pressure at 53.5%).
- **Network Scalability**: In the partially-controlled 3×3 grid, PPO maintains a strong lead over Fixed-Time, narrowing the gap to Max Pressure (−6.4%).

Conclusions & Key Contributions

- **Addressing Benchmark Limitations**: While traditional heuristics like *Max Pressure* and *SOTL* are vehicle-centric and lack pedestrian safety, and state-of-the-art *SPRe+* guarantees zero violations but suffers from high computation cost (QP projection overhead), our approach bridges these gaps.
- **Optimal Safety–Efficiency Trade-off**: The proposed **Action-constrained PPO** achieves the lowest waiting time among safe agents with zero pedestrian-safety violations guaranteed by construction.
- **Eliminated Computational Overhead**: By enforcing feasible-action masking during training, our method completely eliminates the post-hoc projection overhead of SPRe+ and the resampling loops of DQN-AR, ensuring scalability for larger networks.
- **Scalability & Future Horizons**: Longer training is expected to further widen the performance gap in complex 3×3 grids. Future steps include full 9-intersection multi-agent control and evaluation on real-world empirical data.

References

[1] P. Varaiya, "Max pressure control of a network of signalized intersections," *Transportation Research Part C*, vol. 36, pp. 177–195, 2013.

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[3] Y.-H. Hung et al., "SPRe+: Safe policy with repair for action-constrained reinforcement learning," 2025.